



STM32F3 Technical Training

For reference only

Refer to the latest documents for details



CAN Peripheral



CAN Features (1/2)

3

- Supports CAN protocol version 2.0 A, B Active
- Bit rates up to 1Mbit/s
- Transmission
 - Three transmit mailboxes
 - Configurable transmit priority
 - Time Stamp on SOF transmission
- Reception
 - Two receive FIFOs with three stages
 - 14 scalable filter banks
 - Configurable FIFO overrun
 - Time Stamp on SOF reception

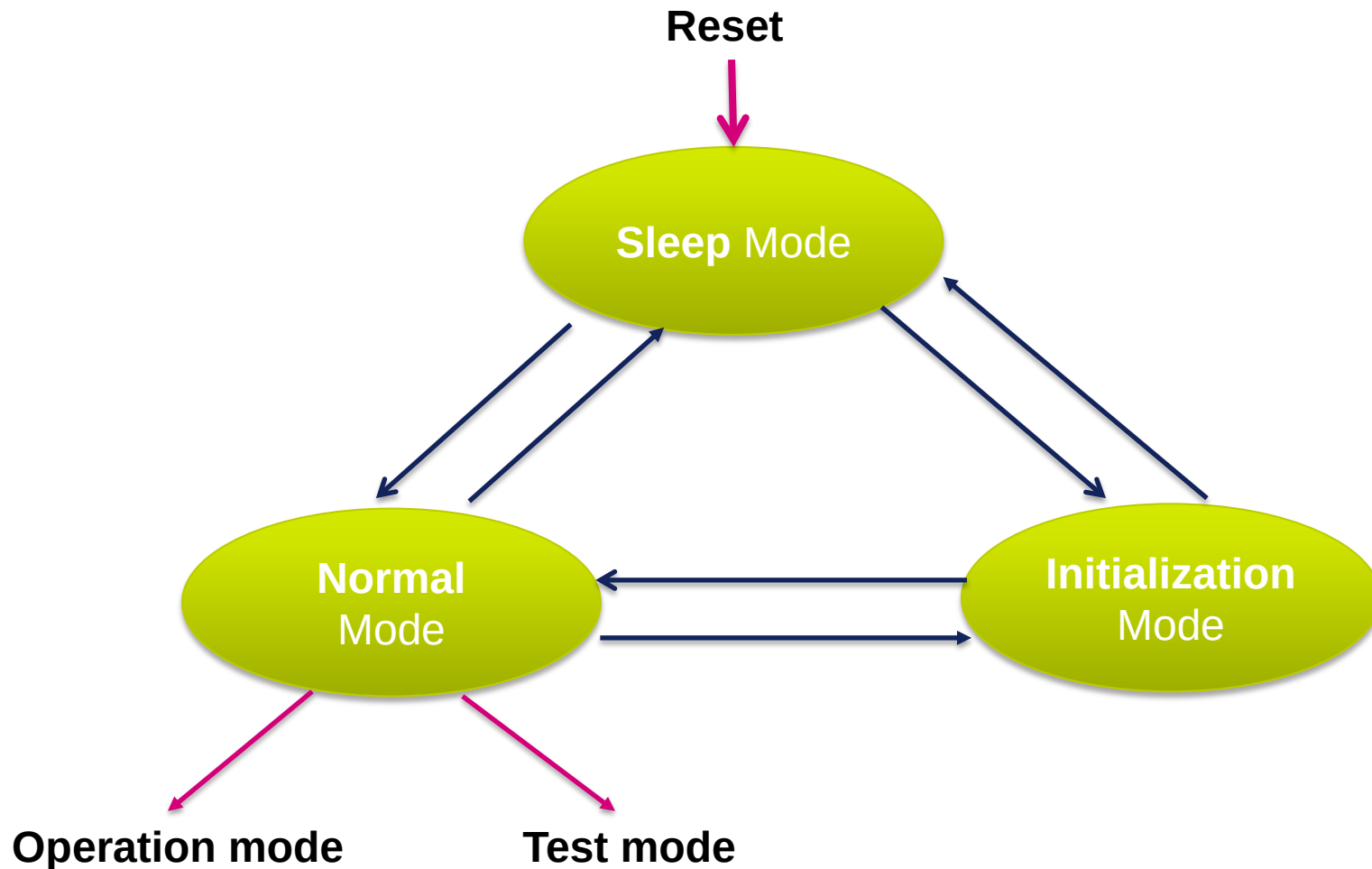
CAN Features (2/2)

4

- Time Triggered Communication option
 - Disable automatic retransmission mode
 - 16-bit free running timer
 - Time Stamp sent in last two data bytes
- Management
 - Maskable interrupts
 - 512 bytes reserved RAM size **(No longer shared with USB)**
 - 4 dedicated interrupt vectors: transmit interrupt, FIFO0 interrupt, FIFO1 interrupt and status change error interrupt

BxCAN operating modes

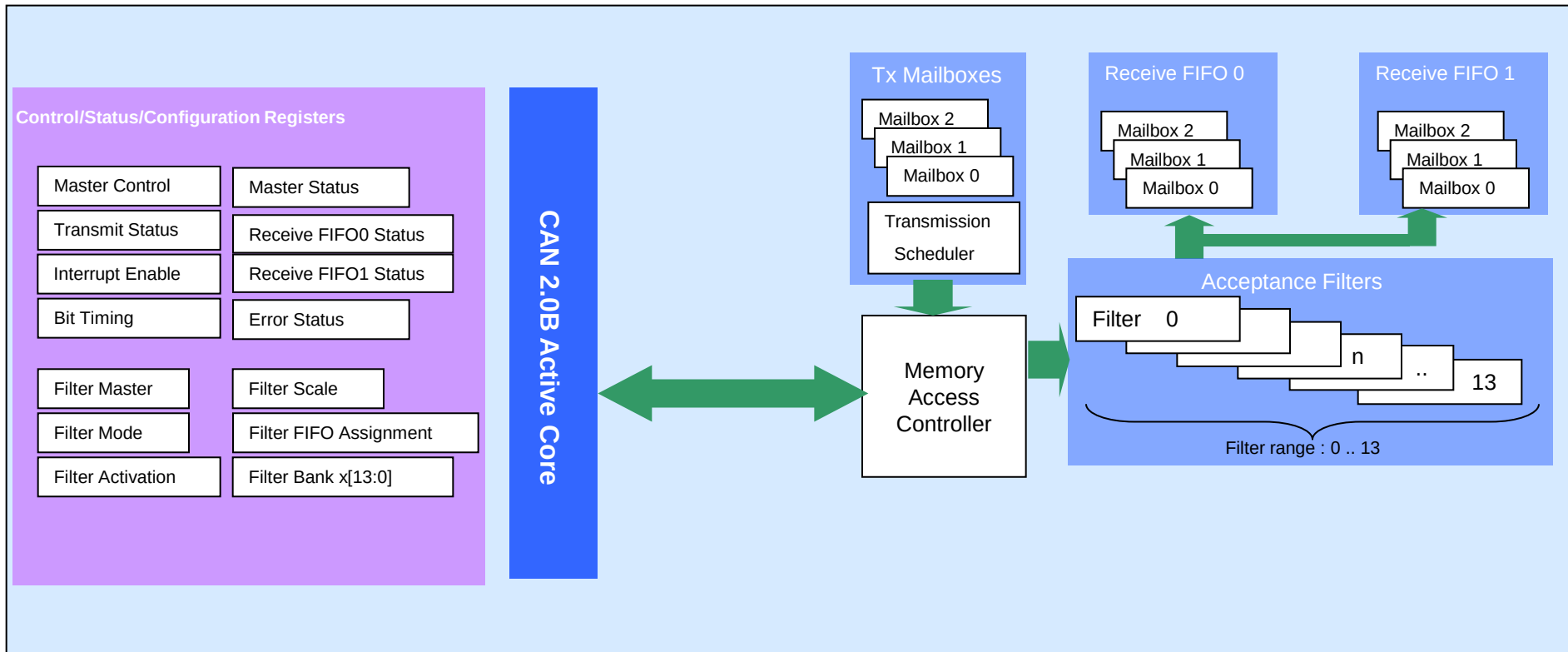
5



- Silent mode
- LoopBack mode
- Loop back combined with silent mode

Block Diagram – BxCAN

6



Same as STM32F10x product

- How many transmit mailboxes are in the STM32F3xxx bxCAN?

- How many operating modes are in the STM32F3xxx bxCAN?

- What is the difference between STM32F3xxx bxCAN and STM32F10x bxCAN?



Thank you !

CONFIDENTIALITY OBLIGATIONS:

THIS DOCUMENT CONTAINS SENSITIVE INFORMATION.

IT IS CLASSIFIED "MICROCONTROLLERS, MEMORIES & SECURE MCUs (MMS) RESTRICTED AND ITS DISTRIBUTION IS SUBMITTED TO ST/MMS AUTHORIZATION

AT ALL TIMES YOU SHOULD COMPLY WITH THE FOLLOWING SECURITY RULES:

- DO NOT COPY OR REPRODUCE ALL OR PART OF THIS DOCUMENT
- KEEP THIS DOCUMENT LOCKED AWAY
- FURTHER COPIES CAN BE PROVIDED ON A "NEED TO KNOW BASIS", PLEASE CONTACT YOUR LOCAL ST SALES OFFICE OR DOCUMENT WRITTER.

Information furnished is believed to be accurate and reliable. However, STMicroelectronics assumes no responsibility for the consequences of use of such information nor for any infringement of patents or other rights of third parties which may result from its use. No license is granted by implication or otherwise under any patent or patent rights of STMicroelectronics. Specifications mentioned in this publication are subject to change without notice. This publication supersedes and replaces all information previously supplied. STMicroelectronics products are not authorized for use as critical components in life support devices or systems without express written approval of STMicroelectronics.

The ST logo is registered trademark of STMicroelectronics
All other names are the property of their respective owners

© 2015 STMicroelectronics - All Rights Reserved

STMicroelectronics group of companies Australia - Brazil - Canada - China - Finland - France - Germany - Hong Kong - India - Israel - Italy - Japan - Malaysia - Malta - Morocco - Singapore - Spain - Sweden - Switzerland - United Kingdom - United States.

www.st.com